MAIS PAC

IPhO 03.03: Data Analysis and Linear Fitting · Worked Solutions

Data Analysis and Linear Fitting: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Solution 1 — Linearising a spring. Return →

(a) Square the theory: $T^2 = (4\pi^2)/(k)m$, so plot T^2 against m; the gradient is $4\pi^2/k$.

(b) k = $4\pi^2/0.82$ = 48 {N,m^(-1)}.

(c) The spring itself has mass, and the clamp and hook contribute a small fixed mass; the effective mass is m + m_({eff}), so $T^2 = (4\pi^2/k)(m+m_{\{eff\}})$ and the intercept is positive. **Check:** the intercept divided by the gradient gives m_({eff}), typically a few grams — a nice extra result from the same graph.

Solution 2 — A log–log plot. Return →

(a) $s = (1)/(2)at^2$ is a power law s = Kt^n with n = 2 and K = a/2.

(b) The log–log gradient is the exponent: 2.

(c) The intercept is log (a/2), so a = 2×10^({intercept}). **Check:** if the data gave a gradient of 1.94 rather than 2, the residuals and the gradient would both suggest either air resistance (which reduces the effective exponent slightly at large t) or timing error.

Solution 3 — Least squares by hand. Return →

(a) x = 2.5, y = 5.0.

(b) Deviations: x- x: -1.5, -0.5, 0.5, 1.5; y- y: -2.9, -1.1, 1.2, 2.8. Then

$$\Sigma\,(x-\,x)(y-\,y) = 4.35+0.55+0.60+4.20 = 9.70,\ \Sigma\,(x-\,x)^2 = 2.25+0.25+0.25+2.25 = 5.0, \tag{1}$$

so m = 9.70/5.0 = 1.94.

(c) c = y - m x = 5.0 - 4.85 = 0.15. The scatter of the points is about ±0.1, so an intercept of 0.15 is marginally significant and worth investigating (a systematic offset in the measuring instrument is the usual culprit). **Check:** the gradient 1.94 is close to 2, which is the value the data were constructed to suggest.

Solution 4 — Uncertainty in a gradient. Return →

(a) Δ m = √((0.036)/((4-2)(5.0))) = √(0.0036) = 0.060.

(b) 0.060/1.94 = 3.1%.

(c) Widening the range increases Σ(x- x)² quadratically in the span, while adding points increases it only linearly. Four points spread over 0.2–4.0 would be far better than twenty crammed into 1.9–2.1. **Check:** this is the quantitative version of the figure on the span of readings.

Solution 5 — Residuals as a diagnostic. Return →

(a) Residuals positive at both ends and negative in the middle form a **U** shape, which is the signature of fitting a straight line to a curve that is concave upwards: the true relationship is not linear over that range.

(b) For a filament lamp the resistance rises with temperature, so V grows faster than I; plotting R = V/I against I (or against T if measured) exposes the dependence instead of hiding it in a curve.

(c) Over a small range of currents the temperature changes little, the resistance is nearly constant, and the straight line is a good local approximation — which is why Ohm's law works so well for small signals. **Check:** the U-shaped residuals are the classic exam answer; a random scatter would have no such order.

Solution 6 — Interpreting R². Return →

(a) 0.9995 and 0.998 are both excellent — indistinguishable in practice, and the difference certainly does not decide which model is right. 0.87 indicates substantial scatter or a poor model.

(b) A parabola sampled only near its vertex looks almost straight, and a straight-line fit through those points can easily give $R^2 > 0.99$ while the true curvature is real.

(c) R^2 measures how much variance the line explains, not whether the model is correct: it can be high for the wrong functional form over a narrow range, and it says nothing about systematic errors. Always look at the residuals. **Check:** R^2 also rises automatically when points are added, so it is not comparable across datasets of different size.

Solution 7 — Cooling data. Return →

(a) T - 20 = (T_0-20)e^(-kt), so ln (T-20) = ln (T_0-20) - kt: a straight line of gradient -k.

(b) ln (60) - ln (30) = k(480), so k = ln 2/480 = 1.44×10^(-3) {s^(-1)}.

(c) At t = 1200 {s}, excess = 60e^(-1.73) = 10.6 {K}, so T = 30.6°{C}. The intercept is ln 60, which is just the starting excess — it carries no extra physics unless the starting temperature is unknown. **Check:** the half-life of the excess temperature is 480 {s}, matching the two given values — that is what the exponential means.

Solution 8 — Two variables with uncertainties. Return →

(a) $g = 4\pi^2 L/T^2$, and T enters squared, so its fractional uncertainty is doubled, whereas L enters to the first power.

(b) L: 0.002/1.000 = 0.2%. T: 2(0.004/2.006) = 0.40%. The T contribution is about twice the L one, and in quadrature the total is 0.45% — dominated by T but not overwhelmingly.

(c) To balance them, time more oscillations (say 100, reducing the T contribution to 0.08%) and measure L with a metre rule to 1 {mm}: then both contributions are about 0.3%. **Check:** the uncertainty is set by the worse of the two, so improving the already-good variable wastes effort.

Solution 9 — Weighted fitting. Return →

(a) Simple mean: (5.0+8.0)/2 = 6.5.

(b) Weights $1/\sigma^2$: 1.0 and 4.0, so the weighted mean is

$$(5.0(1.0)+8.0(4.0))/(5.0) = (37.0)/(5.0) = 7.4. \quad (2)$$

(c) The second measurement is four times more precise, so it carries four times the weight; the weighted mean sits closer to 8.0, as it should. The uncertainty of the weighted mean is $1/\sqrt{(1.0+4.0)} = 0.45$, so the result is (7.4 ± 0.5). **Check:** the simple mean would claim an uncertainty of about 0.45 too, but from inconsistent data — the weighted version uses the information honestly.

Solution 10 — A complete analysis. Return →

(a) R is proportional to L, so plot R against L; the gradient is R/L. Using the first and last points, (4.26-0.86)/(1.00-0.20) = 4.25 Ω,{m^(-1)}; a least-squares fit through all five gives essentially the same value, about 4.26 Ω,{m^(-1)}, with an uncertainty of about 0.03 Ω,{m^(-1)} from the scatter of the points.

(b) $A = \pi d^2/4 = \pi(0.40\times10^{(-3)})^2/4 = 1.26\times10^{(-7)}$ {m²}, and $\rho = (R/L)A = (4.26)(1.26\times10^{(-7)}) = 5.4\times10^{(-7)}$ Ω,{m}.

(c) Fractional uncertainties: gradient 0.03/4.26 = 0.7%, diameter 2(0.01/0.40) = 5.0%. The diameter dominates completely, so the gradient measurement is already good enough and the diameter needs a micrometer used at several points along the wire. **Check:** the result is close to nichrome, and the dominant uncertainty is a diameter measurement that takes ten seconds to improve — the classic practical-exam insight.